

Companion XV: The Early-Universe Gravitational-Wave Spectrum in the Relaxation-Driven Cyclic Cosmology (RDCC) Relaxation-Induced Tensor Damping, Sectoral Asymmetry, and CPT-Sector GW Signatures

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Abstract

The early-Universe gravitational-wave (GW) spectrum provides a uniquely sensitive probe of the microphysical processes that shaped the primordial plasma. In the Relaxation-Driven Cyclic Cosmology (RDCC), the universal relaxation mechanism introduced in Companion X modifies the evolution of tensor perturbations during the earliest stages of cosmic history. The relaxation rate $\Gamma(a) = \alpha(a)H(a)$ introduces an additional damping term in the tensor-mode equation of motion, suppressing the amplitude of primordial gravitational waves on scales that re-entered the horizon during the relaxation-dominated epoch.

Combined with the sectoral temperature asymmetry between the two CPT-conjugate sectors, this leads to a characteristic sectoral GW asymmetry and a relaxation-imprinted suppression feature in the stochastic gravitational-wave background (SGWB). The colder CPT sector experiences weaker damping and produces a distinct tensor spectrum, while the visible sector exhibits a relaxation-induced suppression at high frequencies.

In this Companion we derive the relaxation-corrected tensor-mode evolution, compute the resulting GW transfer function, and obtain the full early-Universe GW spectrum in RDCC. We incorporate the modernized sectoral thermodynamics from Companion XIII, the CPT-sector PBH interpretation from Companion XIV, and the tensor-modulation structure from Companion II. The results establish the SGWB as a central observational probe of the relaxation-driven dynamics that define the RDCC framework.

1 Motivation

The stochastic gravitational-wave background (SGWB) generated in the early Universe is a uniquely sensitive probe of high-energy physics. In standard cosmology, the SGWB is determined by the primordial tensor spectrum and its subsequent evolution through radiation and matter domination. The resulting spectrum is smooth, nearly scale-invariant, and free of dynamical features except for small steps associated with changes in the relativistic degrees of freedom.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the situation is fundamentally different. The universal relaxation mechanism introduced in Companion X modifies the evolution of super-horizon tensor modes during the earliest stages of cosmic history. When the relaxation rate satisfies

$$\Gamma(a) = \alpha(a)H(a) \gtrsim H(a), \tag{1}$$

tensor modes experience an additional damping term that suppresses their amplitude before horizon re-entry. This effect leaves a characteristic imprint on the SGWB: a smooth, power-law suppression at frequencies corresponding to modes that re-entered during the relaxation-dominated epoch.

A second key ingredient is the sectoral temperature asymmetry between the two CPT-conjugate sectors. As shown in Companion X, the temperature ratio

$$\frac{T_B}{T_A} \approx 0.5 \quad (2)$$

implies different relaxation histories, horizon masses, and damping scales in the two sectors. Tensor modes in the colder CPT sector experience weaker damping, leading to a sectoral GW asymmetry.

This Companion develops the early-Universe gravitational-wave sector of RDCC, incorporating the modernized sectoral thermodynamics from Companion XIII, the CPT-sector PBH interpretation from Companion XIV, and the tensor-modulation structure from Companion II. The SGWB emerges as a central observational probe of the relaxation-driven dynamics that define the RDCC framework.

2 Tensor Modes in Standard Cosmology

In standard cosmology, primordial tensor perturbations obey the equation

$$\ddot{h}_k + 3H\dot{h}_k + \frac{k^2}{a^2}h_k = 0, \quad (3)$$

with two distinct regimes:

- **Super-horizon** ($k < aH$): Tensor modes freeze out with constant amplitude.
- **Sub-horizon** ($k > aH$): Tensor modes oscillate with frequency k/a and decay as a^{-1} due to Hubble friction.

The primordial tensor spectrum is parameterized as

$$\mathcal{P}_h(k) = A_t \left(\frac{k}{k_*} \right)^{n_t}, \quad (4)$$

and the present-day GW energy density is

$$\Omega_{\text{GW}}(k) = \frac{1}{12} \left(\frac{k}{a_0 H_0} \right)^2 \mathcal{P}_h(k) T^2(k), \quad (5)$$

where $T(k)$ is the transfer function.

Crucially, standard cosmology contains no additional damping beyond Hubble friction. The SGWB is therefore smooth and featureless, making it exceptionally sensitive to any new early-Universe physics.

3 Tensor Modes in RDCC: Relaxation-Induced Damping

In RDCC, the universal relaxation mechanism modifies the tensor-mode equation by introducing an additional damping term. The evolution equation becomes

$$\ddot{h}_k + (3H + \Gamma(a))\dot{h}_k + \frac{k^2}{a^2}h_k = 0, \quad (6)$$

with

$$\Gamma(a) = \alpha(a)H(a), \quad (7)$$

where $\alpha(a)$ is the relaxation coefficient derived in Companion X.

3.1 Super-Horizon Evolution

For $k < aH$, the equation reduces to

$$\ddot{h}_k + (3 + \alpha)H\dot{h}_k \approx 0. \quad (8)$$

During radiation domination ($H \propto a^{-2}$), the solution is

$$h_k(a) \propto a^{-\alpha/2}. \quad (9)$$

Thus, super-horizon tensor modes are suppressed by

$$S(a) = \left(\frac{a}{a_{\text{in}}}\right)^{-\alpha/2}, \quad (10)$$

where a_{in} marks the onset of the relaxation-dominated epoch.

3.2 Sub-Horizon Evolution

After horizon re-entry ($k > aH$), the equation becomes

$$\ddot{h}_k + (3H + \Gamma)\dot{h}_k + \frac{k^2}{a^2}h_k = 0. \quad (11)$$

The amplitude decays as

$$h_k(a) \propto a^{-1} \exp \left[-\frac{1}{2} \int_{a_{\text{re}}}^a \alpha(a') \frac{da'}{a'} \right]. \quad (12)$$

3.3 Connection to Tensor Modulation (Companion II)

Companion II shows that the same inter-sectoral relaxation dynamics induce a modulation of the tensor sector, producing log-periodic features in the primordial tensor spectrum. The relaxation-induced damping derived here is the *late-time* manifestation of the same mechanism. Both effects originate from the bipartite CPT structure and the α -controlled inter-sectoral coupling.

3.4 Sectoral Asymmetry

Because the CPT-conjugate sector is colder,

$$\alpha_B(a) < \alpha_A(a), \quad (13)$$

tensor modes in sector B experience weaker damping:

$$h_k^{(B)} > h_k^{(A)}. \quad (14)$$

This sectoral GW asymmetry parallels the sectoral PBH asymmetry derived in Companion XIV and is a direct consequence of the RDCC relaxation dynamics.

In the next section, we compute the relaxation-corrected transfer function and derive the full SGWB spectrum.

4 Relaxation-Corrected Transfer Function

The transfer function $T(k)$ encodes the evolution of tensor modes from their primordial generation to the present day. In standard cosmology, $T(k)$ is determined by horizon re-entry, changes in the relativistic degrees of freedom, and the transition from radiation to matter domination. In the Relaxation-Driven Cyclic Cosmology (RDCC), the universal relaxation mechanism introduces an additional suppression factor that modifies the transfer function on scales that re-entered the horizon during the relaxation-dominated epoch.

4.1 Definition

The present-day tensor power spectrum is

$$\mathcal{P}_h(k, a_0) = \mathcal{P}_h(k, a_{\text{prim}}) T^2(k), \quad (15)$$

where $\mathcal{P}_h(k, a_{\text{prim}})$ is the primordial tensor spectrum and $T(k)$ encodes all subsequent evolution.

In RDCC, the transfer function factorizes as

$$T(k) = T_{\text{std}}(k) T_{\text{rel}}(k), \quad (16)$$

where $T_{\text{std}}(k)$ is the standard transfer function and $T_{\text{rel}}(k)$ captures the relaxation-induced suppression.

4.2 Relaxation-Induced Suppression

From the tensor-mode evolution derived in Section 3, the amplitude of a tensor mode is suppressed by two effects:

- **Super-horizon suppression** by the factor $S(a)$,
- **Enhanced sub-horizon damping** due to the additional friction term $\Gamma(a)$.

The combined suppression factor is

$$T_{\text{rel}}(k) = S(a_{\text{re}}) \exp \left[-\frac{1}{2} \int_{a_{\text{re}}}^{a_{\text{end}}} \alpha(a) \frac{da}{a} \right], \quad (17)$$

where a_{re} is the scale factor at horizon re-entry and a_{end} marks the end of the relaxation-dominated epoch.

Using $\Gamma(a) = \alpha(a)H(a)$, the integral simplifies to

$$T_{\text{rel}}(k) \simeq S(a_{\text{re}}) \left(\frac{a_{\text{re}}}{a_{\text{end}}} \right)^{\alpha/2}, \quad (18)$$

if $\alpha(a)$ varies slowly.

4.3 Scale Dependence

Relaxation affects only modes that re-entered the horizon while $\Gamma(a) \geq H(a)$. Defining the characteristic scale k_{rel} by

$$k_{\text{rel}} = a_{\text{rel}} H(a_{\text{rel}}), \quad (19)$$

where a_{rel} satisfies $\Gamma(a_{\text{rel}}) = H(a_{\text{rel}})$, we obtain:

- $k \ll k_{\text{rel}}$: Modes re-enter after relaxation ends. No suppression: $T_{\text{rel}}(k) \approx 1$.
- $k \gg k_{\text{rel}}$: Modes re-enter during relaxation. Strong suppression: $T_{\text{rel}}(k) \propto k^{-\alpha/2}$.

Thus, the relaxation mechanism introduces a power-law suppression feature in the SGWB spectrum at high frequencies.

4.4 Sectoral Asymmetry

Because the two CPT-conjugate sectors have different temperatures,

$$\alpha_B(a) < \alpha_A(a), \quad (20)$$

their relaxation histories differ. The colder CPT sector experiences weaker damping:

$$T_{\text{rel}}^{(B)}(k) > T_{\text{rel}}^{(A)}(k). \quad (21)$$

This sectoral GW asymmetry parallels the sectoral PBH asymmetry derived in Companion XIV and is a direct consequence of the RDCC relaxation dynamics.

4.5 Resulting Transfer Function

Combining the standard and relaxation-induced contributions yields

$$T(k) = T_{\text{std}}(k) S(a_{\text{re}}) \exp \left[-\frac{1}{2} \int_{a_{\text{re}}}^{a_{\text{end}}} \alpha(a) \frac{da}{a} \right]. \quad (22)$$

This transfer function forms the basis for computing the full stochastic gravitational-wave background in RDCC.

5 The Early-Universe GW Spectrum in RDCC

With the relaxation-corrected transfer function derived in Section 4, we now compute the full stochastic gravitational-wave background (SGWB) in the Relaxation-Driven Cyclic Cosmology (RDCC). The resulting spectrum exhibits a characteristic suppression feature at high frequencies, reflecting the damping of tensor modes during the relaxation-dominated epoch.

5.1 General Expression

The present-day GW energy density per logarithmic interval in wavenumber is

$$\Omega_{\text{GW}}(k) = \frac{1}{12} \left(\frac{k}{a_0 H_0} \right)^2 \mathcal{P}_h(k) T^2(k), \quad (23)$$

with

$$T(k) = T_{\text{std}}(k) T_{\text{rel}}(k). \quad (24)$$

The primordial spectrum is parameterized as

$$\mathcal{P}_h(k) = A_t \left(\frac{k}{k_*} \right)^{n_t}. \quad (25)$$

5.2 Relaxation-Induced Suppression Feature

Using the relaxation-corrected transfer function, the GW spectrum becomes

$$\Omega_{\text{GW}}(k) = \Omega_{\text{std}}(k) T_{\text{rel}}^2(k), \quad (26)$$

with

$$T_{\text{rel}}(k) \approx \begin{cases} 1, & k \ll k_{\text{rel}}, \\ \left(\frac{k}{k_{\text{rel}}} \right)^{-\alpha/2}, & k \gg k_{\text{rel}}. \end{cases} \quad (27)$$

Thus, the relaxation mechanism introduces a power-law suppression at high frequencies:

$$\Omega_{\text{GW}}(k) \propto k^{n_t - \alpha}. \quad (28)$$

5.3 Frequency Range of the Suppression Feature

The characteristic frequency associated with k_{rel} is

$$f_{\text{rel}} = \frac{k_{\text{rel}}}{2\pi a_0}. \quad (29)$$

For typical RDCC parameters, the relaxation-dominated epoch occurs at temperatures

$$T \sim 10^7 - 10^{10} \text{ GeV}, \quad (30)$$

corresponding to frequencies

$$f_{\text{rel}} \sim 10^{-3} - 10^2 \text{ Hz}. \quad (31)$$

This range overlaps with the sensitivity windows of LISA, DECIGO, and BBO.

5.4 Sectoral GW Asymmetry

Because $\alpha_B < \alpha_A$, the colder CPT sector has a shallower suppression slope:

$$\Omega_{\text{GW}}^{(B)}(k) > \Omega_{\text{GW}}^{(A)}(k) \quad (32)$$

for $k \gg k_{\text{rel}}$.

5.5 Full RDCC GW Spectrum

Combining all contributions, the RDCC GW spectrum takes the form

$$\Omega_{\text{GW}}(k) = \frac{1}{12} \left(\frac{k}{a_0 H_0} \right)^2 A_t \left(\frac{k}{k_*} \right)^{n_t} T_{\text{std}}^2(k) T_{\text{rel}}^2(k), \quad (33)$$

with

$$T_{\text{rel}}(k) \approx \begin{cases} 1, & k \ll k_{\text{rel}}, \\ \left(\frac{k}{k_{\text{rel}}} \right)^{-\alpha/2}, & k \gg k_{\text{rel}}. \end{cases} \quad (34)$$

This spectrum exhibits a smooth, nearly scale-invariant low-frequency regime and a relaxation-induced suppression at high frequencies. The location and slope of the suppression feature provide direct information about the relaxation dynamics and the temperature asymmetry between the two sectors.

6 Observational Predictions

The relaxation-corrected gravitational-wave (GW) spectrum derived in Section 5 leads to a set of concrete and testable predictions. These predictions arise from the interplay between relaxation-induced tensor damping, the sectoral temperature asymmetry, and the frequency dependence of horizon re-entry. Together, they provide a powerful observational handle on the relaxation-driven dynamics that define the RDCC framework.

6.1 Prediction 1: High-Frequency Suppression Feature

The universal relaxation mechanism introduces a characteristic suppression feature in the stochastic gravitational-wave background (SGWB) at frequencies

$$f \gtrsim f_{\text{rel}} \sim 10^{-3} - 10^2 \text{ Hz}. \quad (35)$$

Modes that re-entered the horizon during the relaxation-dominated epoch experience a power-law suppression,

$$\Omega_{\text{GW}}(f) \propto f^{n_t - \alpha}, \quad (36)$$

with α the relaxation coefficient. This feature is absent in standard cosmology and provides a direct probe of relaxation-driven early-Universe physics.

6.2 Prediction 2: Observable Break Scale

The break frequency f_{rel} marks the transition between the unsuppressed and suppressed regimes. For typical RDCC parameters, f_{rel} lies within the sensitivity windows of LISA, DECIGO, and BBO. A detection of a spectral break at this frequency would provide strong evidence for the relaxation mechanism.

6.3 Prediction 3: Modified Spectral Slope

Above the break scale, the GW spectrum exhibits a modified slope,

$$\frac{d \ln \Omega_{\text{GW}}}{d \ln f} = n_t - \alpha, \quad (37)$$

where $\alpha = \mathcal{O}(0.1 - 1)$ during the relaxation epoch. This leads to a significant deviation from the nearly scale-invariant spectrum predicted by standard cosmology.

6.4 Prediction 4: Sectoral GW Asymmetry

Because the two CPT-conjugate sectors have different temperatures,

$$\alpha_B < \alpha_A, \quad (38)$$

their relaxation histories differ. This implies a sectoral asymmetry in the GW spectrum,

$$\Omega_{\text{GW}}^{(B)}(f) > \Omega_{\text{GW}}^{(A)}(f) \quad (39)$$

for $f \gtrsim f_{\text{rel}}$. While only the visible-sector spectrum is directly observable, the mirror-sector spectrum affects the global energy budget and provides an internal consistency check of the RDCC framework.

6.5 Prediction 5: Consistency with PBH Constraints

The relaxation-induced suppression of tensor modes is consistent with the suppression of primordial black hole (PBH) formation derived in Companion XIV. Both effects originate from the same relaxation dynamics and occur at similar energy scales. A joint analysis of PBH constraints and GW observations can therefore test the internal coherence of RDCC.

6.6 Prediction 6: Distinguishability from Other New-Physics Scenarios

The relaxation-induced suppression feature differs qualitatively from other mechanisms that modify the SGWB, such as:

- changes in relativistic degrees of freedom (g_*),
- phase transitions,
- cosmic strings,
- reheating effects.

These scenarios typically produce sharp features, bumps, or oscillations, whereas RDCC predicts a smooth power-law suppression with a well-defined break scale. This distinct spectral shape allows the relaxation mechanism to be distinguished from other early-Universe physics.

6.7 Prediction 7: Direct Testability with Future Detectors

The predicted suppression feature lies in the frequency range accessible to upcoming space-based interferometers. In particular:

- LISA can probe the low-frequency edge of the suppression feature,
- DECIGO and BBO can probe the full suppression regime,
- high-frequency detectors (MHz–GHz) can test the extreme tail of the spectrum.

A detection of the predicted spectral break and modified slope would provide strong evidence for the relaxation mechanism and the bipartite structure of the early Universe.

7 Conclusion

In this Companion we have developed the gravitational-wave (GW) sector of the Relaxation-Driven Cyclic Cosmology (RDCC), incorporating the modernized relaxation dynamics from Companion X, the sectoral thermodynamics from Companion XIII, the CPT-sector PBH interpretation from Companion XIV, and the tensor-modulation structure from Companion II.

The key result is that the relaxation mechanism introduces a characteristic suppression feature in the GW spectrum at frequencies corresponding to modes that re-entered the horizon during the relaxation-dominated epoch. This suppression arises from two effects: the relaxation-induced damping of super-horizon tensor modes and the enhanced friction term affecting sub-horizon evolution. The resulting spectrum exhibits a smooth power-law suppression above a well-defined break scale, providing a clear observational signature of relaxation-driven early-Universe physics.

A second important result is the sectoral GW asymmetry. Because the two CPT-conjugate sectors have different temperatures, their relaxation histories differ, leading to distinct GW spectra. This sectoral asymmetry parallels the sectoral PBH asymmetry developed in Companion XIV and provides an internal consistency check of the RDCC framework.

With the development of the GW sector, RDCC now provides a coherent and internally consistent description of scalar, tensor, and PBH-related early-Universe physics. The SGWB emerges as a central observational probe of the relaxation-driven dynamics that define the RDCC cosmology.

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